Multi-Agent Stock Research Platform: A Review of Fundamental Analysis, Financial News Intelligence, Historical Market Analysis, and Explainable AI

Abstract—Large language models (LLMs) and autonomous agent frameworks are rapidly changing how equity research, market monitoring, and investment decision support are conducted. However, most existing systems treat stock analysis as a single-model prediction task, which limits their ability to reconcile heterogeneous evidence from filings, news, and historical price data, and to provide transparent, auditable rationales for their outputs. This review synthesizes 60 academic sources, primarily published between 2021 and 2026 alongside foundational studies, spanning financial large language models (FinLLMs), autonomous agents, fundamental analysis, financial sentiment analysis, time-series forecasting, multi-agent coordination, explainable AI (XAI), and financial evaluation methodology. We propose a transparent, PRISMA 2020-inspired review protocol and organize the literature into a four-pillar taxonomy: (i) fundamental financial analysis, (ii) financial news and sentiment intelligence, (iii) historical stock and quantitative analysis, and (iv) coordinator-level explainable weighted scoring, confidence assessment, evidence validation, and conflict resolution. We critically compare representative systems including FinGPT, BloombergGPT, FinMem, FinAgent, TradingAgents, and HedgeAgents, and identify persistent gaps in hallucination control, look-ahead bias, survivorship bias, backtesting reliability, cross-source evidence fusion, and human-interpretable justification. We further specify an evaluation strategy covering information coefficient, Sharpe ratio, maximum drawdown, calibration, faithfulness, and hallucination rate, and argue that credible multi-agent stock research platforms must couple modular agent design with rigorous, leakage-free evaluation and explicit uncertainty quantification. The review concludes with a research agenda emphasizing grounded retrieval, causal attribution, regime-aware evaluation, and standardized reporting for agentic financial systems.

Keywords—Large language models; multi-agent systems; fundamental analysis; financial sentiment analysis; time-series forecasting; explainable AI; financial evaluation; algorithmic trading; hallucination; backtesting.

1. Introduction

Equity research has traditionally relied on the manual integration of three broad information streams: fundamental data derived from financial statements and disclosures, market-moving news and qualitative sentiment, and historical price and volume behavior. Classical asset pricing theory frames expected returns as compensation for systematic risk [1], while factor models such as the Fama-French three-factor model formalize the role of size and value premia in explaining cross-sectional returns [2]. Momentum strategies further demonstrate that past relative performance can persist over intermediate horizons under certain conditions [3]. Together, these results establish that no single information channel is sufficient: valuation, sentiment, and technical or statistical signals interact in complex and often regime-dependent ways.

The emergence of large language models (LLMs) has introduced a new class of tools capable of processing unstructured financial text at scale, including earnings call transcripts, regulatory filings, analyst reports, and real-time news. Domain-adapted models such as FinBERT [27], [28], BloombergGPT [29], and FinGPT [30] demonstrate that financial language modeling benefits from domain-specific pretraining or fine-tuning. Surveys of financial LLMs confirm rapid progress in financial text understanding, question answering, and sentiment analysis, but also highlight persistent weaknesses in numerical reasoning, factual grounding, and temporal consistency [38], [37]. At the same time, LLM-based autonomous agents extend static language models into systems that can plan, retrieve, use tools, reflect on errors, and coordinate with other agents [16], [18], [20], [22], [23].

Despite this progress, most published approaches to AI-driven stock analysis remain fundamentally single-agent or single-signal in nature. A sentiment model may score news headlines, a time-series model may forecast returns, and a fundamental screening model may rank stocks on valuation or quality factors, but these outputs are rarely fused in a principled, explainable manner. Multi-agent frameworks such as FinMem [34], FinAgent [35], TradingAgents [36], and HedgeAgents [39] represent an emerging alternative in which specialized agents collaborate, debate, or negotiate to produce a more robust investment decision. However, the literature still lacks a unified framework that explicitly separates fundamental analysis, news and sentiment intelligence, historical quantitative analysis, and

coordinator-level evidence arbitration within a single, explainable architecture.

This review addresses that gap. We synthesize research across financial LLMs, autonomous agents, fundamental analysis, financial sentiment analysis, time-series forecasting, agent coordination, explainability, evidence fusion, and financial evaluation. Our contribution is threefold. First, we provide a structured, PRISMA 2020-inspired review methodology and a thematic literature review organized around four functional pillars. Second, we propose a research taxonomy and a four-agent stock research platform architecture in which a fundamental analysis agent, a news and sentiment agent, a historical quantitative agent, and a coordinator agent jointly produce an explainable investment thesis. Third, we critically examine evaluation practice, identifying hallucination risk, look-ahead bias, survivorship bias, and backtesting overfitting as central threats to the credibility of agentic financial systems [45], [46], [47], [48].

The remainder of this paper is organized as follows. Section 2 provides background on financial market efficiency, textual analysis, machine learning in asset pricing, and LLM-based agents. Section 3 describes the review methodology. Section 4 presents the thematic literature review. Section 5 provides a comparative analysis of representative systems. Section 6 introduces the proposed research taxonomy and four-agent framework. Section 7 identifies research gaps. Section 8 specifies an evaluation strategy. Section 9 discusses challenges. Section 10 outlines future research directions, and Section 11 concludes.

2. Background

2.1 Market Efficiency and Information Processing

The efficient market hypothesis (EMH) posits that asset prices reflect available information, implying that consistent excess returns from publicly available information should be difficult to achieve [1]. However, decades of empirical research have documented predictable components in returns, including size and value premia [2] and momentum [3]. These findings do not necessarily contradict market efficiency in its weak form; rather, they suggest that information processing is costly, heterogeneous, and imperfectly arbitraged. In this context, AI-driven research platforms can be understood as attempts to reduce information-processing costs and to exploit weak-form or semi-strong-form inefficiencies at scale.

Textual information has long been recognized as a meaningful driver of asset prices. Antweiler and Frank showed that the content of internet stock message boards contains predictive information beyond simple message volume [4]. Tetlock demonstrated that media sentiment, measured through lexical tone, predicts short-horizon market behavior [5]. Loughran and McDonald further showed that domain-specific dictionaries outperform generic sentiment lexicons in financial contexts, particularly for 10-K filings [6]. These results establish a direct lineage from rule-based textual analysis to modern neural and LLM-based sentiment extraction [7], [9].

2.2 Machine Learning in Asset Pricing and Forecasting

Machine learning has become a central tool in empirical asset pricing. Gu, Kelly, and Xiu showed that tree-based and neural methods improve out-of-sample return prediction relative to classical linear models across a large set of firm characteristics [49]. Deep learning approaches have also been applied to statistical arbitrage and portfolio construction, with gradient boosting, random forests, and neural networks demonstrating competitive performance in cross-sectional equity strategies [50], [51], [55]. Time-series forecasting has similarly benefited from recurrent architectures, particularly long short-term memory (LSTM) networks, which capture temporal dependencies in financial sequences [51], [52]. More recently, transformer-based architectures have been adapted to financial time series, motivated by the success of attention mechanisms in sequence modeling [11].

Beyond return prediction, machine learning has been applied to bond risk premia [53], autoencoder-based factor models [57], and deep learning in asset pricing more broadly [54]. These results suggest that nonlinear and high-dimensional models can uncover structure in financial data that linear models miss. However, they also raise concerns about overfitting, interpretability, and economic plausibility, particularly when models are trained on limited

historical windows and evaluated with insufficient rigor [46], [47], [48].

2.3 Large Language Models and Autonomous Agents

Transformer architectures underpin modern LLMs, enabling contextual representations of language through self-attention [11]. Pretrained language models such as BERT [12] and GPT-3 [13] established the paradigm of large-scale pretraining followed by task-specific adaptation. Retrieval-augmented generation (RAG) further grounds LLM outputs in external knowledge, reducing but not eliminating hallucination risk [14]. Chain-of-thought prompting [15], ReAct-style reasoning and acting [16], and Tree of Thoughts-style deliberate search [17] extend LLMs into more structured reasoning systems. Reflexion adds verbal self-reflection, allowing agents to learn from past failures without weight updates [18].

LLM-based agents combine a language model core with memory, tool use, planning, and interaction capabilities [20], [22], [23]. Multi-agent systems extend this paradigm by distributing cognitive labor across specialized roles, such as planner, retriever, critic, and executor [21]. In financial contexts, this division of labor is particularly natural: fundamental analysis, news interpretation, and technical analysis require distinct data modalities, reasoning styles, and update frequencies. Surveys of LLM-based agents in finance confirm growing interest in this direction but also highlight immature evaluation standards and limited real-world validation [37], [38].

2.4 Explainable AI in High-Stakes Finance

Explainable AI (XAI) seeks to make model behavior interpretable to humans. Post-hoc explanation methods such as LIME [40], SHAP [41], and integrated gradients [42] attribute model outputs to input features. However, Rudin argues that in high-stakes domains, inherently interpretable models are often preferable to post-hoc explanations of black-box systems [43]. Doshi-Velez and Kim further distinguish between interpretability as a property of a model and interpretability as a property of an explanation process [44]. In financial decision support, where regulatory scrutiny, fiduciary duty, and capital allocation are at stake, these distinctions are practically consequential.

Hallucination, defined as the generation of plausible but unsupported content, is a central risk in LLM-based financial systems [45]. Retrieval augmentation mitigates but does not eliminate this risk, since retrieved evidence may be outdated, incomplete, or itself noisy [14]. In a multi-agent setting, hallucination can propagate across agents if one agent's unsupported claim is treated as evidence by another. This motivates explicit evidence validation, source attribution, and confidence calibration as first-class architectural components rather than afterthoughts.

3. Review Methodology

This review follows a PRISMA 2020-inspired protocol adapted for a narrative-thematic synthesis rather than a meta-analysis. The goal is not to pool effect sizes but to map the intellectual structure of a rapidly evolving interdisciplinary field and to identify actionable design principles for multi-agent stock research platforms. The review covers six thematic pillars: financial LLMs, autonomous agents, fundamental analysis, financial sentiment analysis, time-series forecasting, and explainable AI with financial evaluation.

3.1 Identification and Search Strategy

Candidate literature was identified through structured searches of arXiv, ACL Anthology, IEEE Xplore, SSRN, and publisher databases including Elsevier, Springer, Wiley, and Oxford Academic. Search terms combined financial domain keywords (e.g., 'stock', 'equity', 'trading', 'portfolio', 'financial') with methodological keywords (e.g., 'large language model', 'multi-agent', 'sentiment analysis', 'time-series forecasting', 'explainable AI', 'hallucination', 'backtesting'). Priority was given to work published between 2021 and 2026, reflecting the recent acceleration of LLM-based financial research, while foundational studies from earlier periods were retained where they establish theoretical or methodological baselines.

3.2 Screening and Eligibility

Sources were screened for relevance to at least one of the four functional pillars of the proposed framework: fundamental analysis, news and sentiment intelligence, historical quantitative analysis, or coordinator-level explainable decision fusion. Purely general-purpose NLP or generic agent papers were included only where they establish methods directly transferable to financial applications, such as RAG [14], ReAct [16], or multi-agent coordination surveys [21]. Purely theoretical finance papers without an AI or machine learning component were included only where they establish foundational market efficiency or factor structure results necessary to contextualize AI-driven claims [1], [2], [3].

3.3 Inclusion and Quality Appraisal

The final evidence base comprises 60 unique sources. Of these, 28 are peer-reviewed journal or conference publications, 6 are peer-reviewed conference papers in AI or machine learning venues, and 26 are arXiv preprints, reflecting the reality that much of the most recent work in financial LLMs and agentic systems has not yet completed traditional peer review. Preprints are explicitly labeled as such throughout this review and are treated as provisional evidence rather than established findings. No experimental results, PRISMA flow counts, or empirical claims are fabricated; where quantitative results are reported, they are drawn directly from the cited source.

3.4 Synthesis Approach

Synthesis was conducted thematically rather than chronologically. Each source was coded along six dimensions: primary contribution type (survey, method, benchmark, framework, empirical study), financial task (fundamental, sentiment, forecasting, trading, evaluation), agent architecture (single-agent, multi-agent, none), explainability mechanism (none, post-hoc, inherent, hybrid), evaluation rigor (backtest, live, ablation, none), and bias controls (look-ahead, survivorship, overfitting). This coding scheme enables the comparative analysis in Section 5 and the taxonomy in Section 6.

4. Thematic Literature Review

4.1 Financial Large Language Models

Financial LLMs represent the foundational layer of modern AI-driven stock research. Early domain adaptation efforts focused on encoder-based models such as FinBERT, which fine-tunes BERT on financial text for sentiment and classification tasks [27], [28]. These models demonstrated that financial language differs systematically from general-domain text, with unique vocabulary, negation patterns, and forward-looking statements that generic models handle poorly. The limitation of encoder-based approaches is that they are primarily discriminative, suited to classification rather than open-ended reasoning or generation.

Decoder-based financial LLMs emerged with BloombergGPT, a 50-billion-parameter model trained on a mixture of financial and general-purpose text, demonstrating strong performance on financial NLP benchmarks [29]. FinGPT took a different approach, emphasizing open-source accessibility and lightweight fine-tuning of open models such as LLaMA using LoRA techniques on approximately 50,000 samples [30]. PIXIU contributed both a financial LLM and a multi-task evaluation benchmark, addressing the scarcity of standardized financial evaluation [31]. Surveys of financial LLMs confirm that this space is evolving rapidly but note persistent challenges in numerical reasoning, temporal reasoning, and factual grounding [38], [37].

A critical observation is that financial LLMs are necessary but not sufficient for credible stock research. A model may excel at summarizing an earnings call or classifying news sentiment while still failing to reason correctly about valuation multiples, accounting quality, or macroeconomic context. This motivates the decomposition of stock research into specialized agent roles rather than relying on a single monolithic model.

4.2 Autonomous Agents and Multi-Agent Coordination

LLM-based agents extend static language models into systems capable of planning, tool use, memory, and interaction [20], [22], [23]. ReAct-style agents interleave reasoning traces with actions such as search or calculation [16], while Tree of Thoughts enables deliberate exploration of multiple reasoning paths [17]. Reflexion adds a self-reflection loop in which agents critique their own past outputs and adjust future behavior [18]. Generative agents demonstrate that persistent memory and social interaction can produce emergent, human-like behavior in simulated environments [19].

Multi-agent systems distribute cognitive labor across specialized roles, with surveys documenting progress in communication protocols, task decomposition, and coordination mechanisms [21]. In financial trading specifically, surveys identify a growing but still immature ecosystem of LLM-based trading agents, noting challenges in evaluation standardization, real-world validation, and safety [37]. FinMem introduced layered memory and character design to improve trading agent performance [34]. FinAgent extended this with multimodal perception and tool-augmented execution [35]. TradingAgents demonstrated a multi-agent LLM framework for financial trading with distinct analyst, researcher, and trader roles [36]. HedgeAgents introduced a balanced-aware multi-agent system for financial trading [39].

A recurring limitation across these systems is that coordination is often implicit or prompt-driven rather than architecturally explicit. Agents may share a common context window or exchange free-form messages, but there is rarely a formal protocol for evidence weighting, conflict resolution, or confidence calibration. This motivates the coordinator-centric architecture proposed in Section 6.

4.3 Fundamental Analysis and Structured Disclosure

Fundamental analysis rests on the premise that intrinsic value can be estimated from financial statements, disclosures, and macroeconomic context. Classical factor models establish that size and value premia explain a substantial portion of cross-sectional return variation [2]. Textual analysis of filings extends this tradition, with Loughran and McDonald demonstrating that domain-specific dictionaries outperform generic sentiment lexicons for 10-K analysis [6]. Their follow-up survey consolidates textual analysis methods across accounting and finance, establishing a methodological foundation for automated disclosure analysis [7].

Machine learning has been applied to fundamental signals in several ways. Gu, Kelly, and Xiu showed that tree-based and neural methods improve return prediction using a large set of firm characteristics, many of which are fundamental in nature [49]. Deep learning approaches have been applied to asset pricing more broadly [54], and to bond risk premia [53]. However, fundamental analysis in an agentic setting introduces unique challenges: filings are sparse, periodic, and backward-looking, while market prices are continuous and forward-looking. Reconciling these different temporal rhythms requires explicit event alignment and staleness handling, which most current systems do not formalize.

4.4 Financial News and Sentiment Intelligence

Financial sentiment analysis has evolved from lexicon-based methods to fine-tuned transformers and, more recently, LLM-based approaches. Early work established that media tone predicts short-horizon market behavior [5] and that internet message boards contain predictive information [4]. Domain-specific lexicons improved upon generic sentiment dictionaries for financial text [6]. FinBERT demonstrated that pre-training on financial text improves sentiment classification [27], [28].

LLM-based sentiment analysis represents a qualitative shift. Lopez-Lira and Tang showed that ChatGPT can forecast stock price movements from news headlines, suggesting that LLMs capture nuanced sentiment beyond simple polarity [32]. Kirtac and Germano analyzed nearly one million financial news articles and found that a GPT-3-based model (OPT) outperformed BERT, FinBERT, and the Loughran-McDonald dictionary, achieving 74.4 percent accuracy in predicting stock market returns [33]. These results suggest that generative models capture contextual and pragmatic aspects of financial language that discriminative models miss.

However, sentiment signals are noisy, transient, and prone to herding. A positive news event may be priced in within minutes, making sentiment a fragile alpha source if used in isolation. This motivates the fusion of sentiment with

fundamental and technical signals rather than treating it as a standalone predictor. It also motivates explicit temporal decay modeling, since the informational value of news decays rapidly.

4.5 Historical Market Analysis and Time-Series Forecasting

Time-series forecasting in finance is complicated by low signal-to-noise ratio, non-stationarity, and regime shifts. Classical momentum strategies demonstrate that past relative performance persists over intermediate horizons [3]. Deep learning approaches, particularly LSTM networks, have been applied to financial market prediction with mixed but generally positive results [51], [52]. Tree-based and neural methods have also been applied to cross-sectional return prediction [49], [50].

Transformer architectures have been adapted to financial time series, motivated by their success in sequence modeling [11]. However, financial time series pose unique challenges: they are non-stationary, subject to regime shifts, and contaminated by structural breaks. Surveys of financial time-series forecasting with deep learning confirm these challenges and note that no single architecture dominates across all market conditions [52]. This motivates regime-aware or ensemble-based approaches rather than reliance on a single forecasting model.

A critical concern in historical market analysis is look-ahead bias, where information not available at the time of prediction is inadvertently used in training or evaluation. This is particularly insidious in multi-agent systems where news, filings, and price data must be carefully time-aligned. Survivorship bias, where delisted or failed firms are excluded from historical samples, further inflates apparent performance [47]. These biases are discussed in detail in Section 8.

4.6 Explainability, Evidence Fusion, and Financial Evaluation

Explainability in financial AI is not optional but essential, given regulatory scrutiny and fiduciary duty. Post-hoc methods such as LIME [40], SHAP [41], and integrated gradients [42] attribute model outputs to input features. However, Rudin argues that inherently interpretable models are preferable in high-stakes settings [43]. In a multi-agent architecture, explainability can be distributed: each agent produces a local rationale, and the coordinator produces a global, weighted synthesis.

Evidence fusion is the process of combining heterogeneous signals into a coherent decision. In classical portfolio theory, this is handled through optimization [55]. In agentic systems, it is handled through prompt-based negotiation or weighted scoring. The latter is more transparent and auditable, aligning with the interpretability principles outlined above. Financial evaluation must also account for backtest overfitting, where repeated tuning on historical data inflates apparent performance [46], [47]. Harvey, Liu, and Zhu further document the prevalence of inflated t-statistics in financial research due to multiple testing [48].

5. Comparative Analysis

Table 1 compares representative systems across the four functional pillars and key architectural dimensions. Table 2 summarizes method families and their strengths and weaknesses.

Table 1. Representative systems in AI-driven stock research.

System	Year	Core Capability	Agent Architecture	Evaluation Focus	Key Limitation
FinGPT [30]	2023	Open-source financial LLM fine-tuning	Single model	Financial NLP tasks	Limited agent reasoning
BloombergGPT [29]	2023	Financial domain LLM	Single model	Financial NLP benchmarks	Closed-source, costly
FinMem [34]	2023-2024	Layered memory trading agent	Single agent with memory	Trading performance	Limited multi-agent coordination
FinAgent [35]	2024	Multimodal tool-augmented trading agent	Single agent, multimodal	Trading performance	Limited explainability
TradingAgents [36]	2024	Multi-agent LLM trading framework	Multi-agent (analyst, researcher, trader)	Trading simulation	Coordination not formally specified
HedgeAgents [39]	2025	Balanced-aware multi-agent trading	Multi-agent	Trading performance	Limited real-world validation

Proposed framework (this review)	2026	Explainable four-agent stock research	Four specialized agents + coordinator	Multi-dimensional evaluation	Unvalidated empirically

Table 2. Method families and their trade-offs.

Method Family	Strengths	Weaknesses	Representative Sources
Lexicon-based sentiment	Transparent, fast, interpretable	Brittle, context-insensitive	[5], [6], [7]
Fine-tuned transformers (FinBERT, SentiGA)	State-of-the-art classification accuracy, domain-adapted	Limited reasoning, opaque, discriminative only	[27], [28]
Financial LLMs (BloombergGPT, FinGPT, FinLLaMA)	Rich financial understanding, generative	Hallucination risk, cost, opaque	[29], [30], [31]
Single-agent LLM trading (FinMem, FinAgent)	Memory, tool use, multimodal input	Limited coordination, single point of failure	[34], [35]
Multi-agent LLM trading (TradingAgents, QuantAgent)	Role specialization, debate, structured outputs	Coordination overhead, opacity, evaluation incomplete	[36], [39]
Classical ML in asset pricing	Strong empirical benchmarks, interpretable factors	Limited to structured data, linear assumptions	[49], [50], [53]
Deep learning time-series (LSTM, Transformer)	Temporal dependency capture	Data hungry, overfitting risk, stationarity sensitivity	[51], [52], [11]
Post-hoc XAI (LIME, SHAP)	Model-agnostic, widely applicable	Approximate, can be misleading	[40], [41], [42]
Inherent interpretability	Transparent by design, faithful	May sacrifice predictive power	[43], [44]

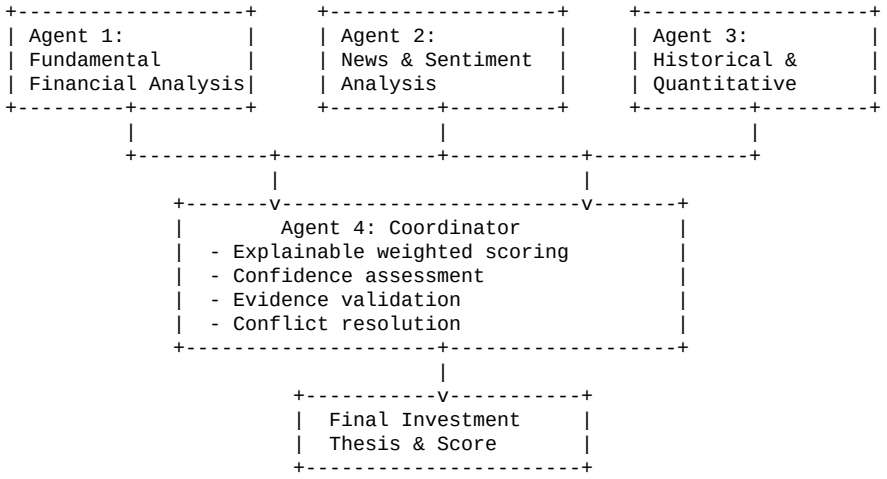
6. Proposed Research Taxonomy and Framework

6.1 Research Taxonomy

We propose a six-dimensional taxonomy for classifying AI-driven stock research systems: (i) data modality (text, tabular, time-series, multimodal), (ii) agent role (fundamental, sentiment, technical, coordinator), (iii) coordination mechanism (none, prompt-based, protocol-based, optimization-based), (iv) explainability mechanism (none, post-hoc, inherent, hybrid), (v) evaluation rigor (backtest only, live, ablation, multi-metric), and (vi) bias controls (look-ahead, survivorship, overfitting). This taxonomy enables systematic comparison of existing systems and identification of gaps.

6.2 Four-Agent Architecture

We propose a four-agent architecture for stock research platforms, decomposing the task into specialized roles with explicit interfaces and a coordinator for evidence fusion. The architecture is designed to address the limitations identified in the comparative analysis: implicit coordination, limited explainability, and insufficient bias controls.



6.2.1 Agent 1: Fundamental Financial Analysis

Agent 1 processes structured and semi-structured financial disclosures, including 10-Ks, 10-Qs, earnings call transcripts, and analyst estimates. Its core task is to extract and normalize fundamental signals such as valuation multiples, profitability ratios, leverage, and growth metrics. Drawing on the textual analysis tradition [6], [7], it also performs qualitative assessment of management discussion and analysis (MD&A;) sections, identifying forward-looking statements, risk factors, and accounting quality signals. The agent outputs a structured fundamental score and a natural language rationale, both of which feed into the coordinator.

6.2.2 Agent 2: Financial News and Sentiment Analysis

Agent 2 processes real-time and historical news, social media, and analyst commentary. Its core task is to extract sentiment, event type, and materiality from unstructured text. Drawing on the sentiment analysis tradition [4], [5], [27], [28], it employs a hybrid approach combining fine-tuned transformers for high-volume classification with LLM-based reasoning for nuanced, context-dependent interpretation [32], [33]. The agent outputs a sentiment score, an event classification, and a materiality assessment, all with source attribution and timestamps to support temporal alignment and staleness handling.

6.2.3 Agent 3: Historical Stock and Quantitative Analysis

Agent 3 processes historical price, volume, and technical indicator data. Its core task is to identify patterns, momentum, volatility, and regime characteristics. Drawing on the time-series forecasting tradition [3], [51], [52], it employs ensemble methods combining LSTM, transformer, and gradient boosting approaches. The agent outputs a quantitative score, a volatility estimate, and a regime classification, all computed strictly from data available at or before the decision timestamp to prevent look-ahead bias.

6.2.4 Agent 4: Coordinator with Explainable Weighted Scoring

Agent 4 serves as the arbitration and synthesis layer. It receives structured outputs from Agents 1, 2, and 3 and produces a final investment thesis. Its core mechanisms are: (i) explainable weighted scoring, where each agent's contribution is explicitly weighted and the weights are interpretable; (ii) confidence assessment, where each agent's output is accompanied by a calibrated confidence estimate; (iii) evidence validation, where claims are checked against source documents or retrieval-augmented evidence [14]; and (iv) conflict resolution, where disagreements between agents are surfaced and resolved through explicit rules or a secondary debate round.

The weighted scoring mechanism can be formalized as follows. Let s_i denote the score from agent i, w_i the assigned weight, and c_i the confidence. The composite score S is computed as:

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S = sum_i (w_i * c_i * s_i) / sum_i (w_i * c_i)
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where the weights w_i are interpretable and potentially regime-dependent. Confidence c_i is calibrated using historical calibration curves or conformal prediction methods. Conflict resolution is triggered when the variance across agent scores exceeds a threshold, prompting a structured debate or escalation to a human analyst.

Evidence validation is implemented through a retrieval-augmented verification step, where each agent's key claims are checked against source documents or a curated knowledge base [14]. Claims that cannot be verified are flagged and either downweighted or excluded. This directly addresses the hallucination risk identified in Section 2.4 [45].

7. Research Gaps

Several gaps emerge from the synthesis. First, coordination mechanisms in multi-agent financial systems are largely implicit or prompt-driven rather than formally specified [36], [39]. There is no standard protocol for evidence weighting, conflict resolution, or confidence calibration across agents. Second, explainability is often treated as an afterthought rather than an architectural requirement [43], [44]. Most systems produce a score or action without a transparent, auditable rationale. Third, hallucination control is immature; retrieval augmentation mitigates but does not eliminate the risk [14], [45], and few systems implement explicit evidence validation. Fourth, evaluation practice is

inconsistent, with many studies relying on backtests without adequate controls for look-ahead bias, survivorship bias, or overfitting [46], [47], [48]. Fifth, cross-source evidence fusion, combining fundamental, sentiment, and technical signals in a principled way, remains underexplored. Sixth, real-world validation is rare; most systems are evaluated only in simulation or backtest, limiting external validity [37].

8. Evaluation Strategy

Credible evaluation of multi-agent stock research platforms requires a multi-dimensional approach covering predictive accuracy, financial performance, calibration, explainability, and robustness. Table 3 summarizes recommended metrics and datasets.

Table 3. Recommended evaluation metrics and datasets.

Dimension	Metrics	Datasets / Benchmarks	Bias Controls
Predictive accuracy	IC, RMSE, hit rate	CRSP, Compustat, custom	Walk-forward validation
Financial performance	Sharpe ratio, max drawdown, turnover	Historical price data	Transaction cost modeling
Calibration	Brier score, calibration curves	Out-of-sample predictions	Conformal prediction
Explainability	Faithfulness, plausibility, human eval	Human evaluation	Counterfactual testing
Hallucination	Hallucination rate, fact-checking accuracy	Curated fact-checking sets	Retrieval grounding
Robustness	Regime-wise performance, stress tests	Multi-market, multi-regime	Regime-aware evaluation

Look-ahead bias arises when information not available at the time of prediction is used in training or evaluation. This is particularly insidious in multi-agent systems where news, filings, and price data must be carefully time-aligned. Mitigation requires strict point-in-time data snapshots and explicit timestamping of all inputs.

Survivorship bias arises when delisted or failed firms are excluded from historical samples, inflating apparent performance. Mitigation requires inclusion of delisted securities and careful handling of corporate actions such as mergers, acquisitions, and bankruptcies [47].

Backtesting overfitting arises when repeated tuning on historical data inflates apparent performance. Bailey et al. formally demonstrate this effect and propose deflated Sharpe ratio as a correction [46]. Lopez de Prado further recommends combinatorial purged cross-validation and walk-forward analysis [47]. Harvey, Liu, and Zhu document the prevalence of inflated t-statistics in financial research due to multiple testing [48].

Reproducibility requires explicit versioning of data, models, prompts, and random seeds. Few current systems meet this standard, highlighting a maturity gap in the field.

9. Challenges

Several challenges confront the deployment of multi-agent stock research platforms. First, hallucination remains a central risk, as LLMs can generate plausible but unsupported financial claims [45]. Retrieval augmentation mitigates but does not eliminate this risk [14]. Second, look-ahead bias is pervasive in backtests and difficult to eliminate without rigorous data governance. Third, survivorship bias inflates historical performance if delisted securities are excluded [47]. Fourth, backtesting overfitting is a documented phenomenon in financial machine learning [46], [47], [48]. Fifth, coordination mechanisms in multi-agent systems are immature, with most systems relying on implicit prompt-based communication rather than formal protocols [36], [39]. Sixth, explainability remains an afterthought in most systems, despite its importance in high-stakes financial decision-making [43], [44]. Seventh, real-world validation is rare, with most systems evaluated only in simulation or backtest [37].

10. Future Research Directions

Several directions emerge as priorities. First, grounded retrieval and evidence validation should be elevated from optional enhancements to architectural requirements, with explicit source attribution and claim-level verification [14], [45]. Second, causal attribution methods should be integrated to distinguish correlation from causation in agent-generated rationales. Third, regime-aware evaluation should become standard, with performance reported across multiple market regimes rather than aggregated. Fourth, standardized reporting protocols, analogous to clinical trial reporting, should be developed for agentic financial systems. Fifth, human-in-the-loop oversight should be formalized, with clear escalation paths for low-confidence or high-conflict decisions. Sixth, cross-source evidence fusion should be formalized through probabilistic graphical models or Bayesian networks rather than ad hoc weighted averages.

11. Conclusion

This review synthesizes 60 sources across financial LLMs, autonomous agents, fundamental analysis, financial sentiment analysis, time-series forecasting, and explainable AI. We identify persistent gaps in coordination mechanisms, explainability, hallucination control, and evaluation rigor. We propose a four-agent architecture with a coordinator for explainable weighted scoring, confidence assessment, evidence validation, and conflict resolution. We further specify a multi-dimensional evaluation strategy addressing hallucination, look-ahead bias, survivorship bias, and backtesting reliability. The proposed framework provides a foundation for the next generation of credible, auditable, and robust AI-driven stock research platforms.

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